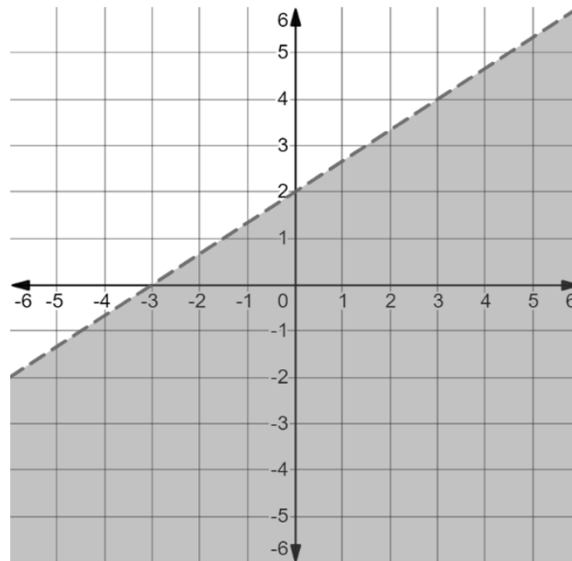


Algebra 1
Unit 6
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

The graph of a two variable linear inequality is shown.



1. Write the inequality of the solution set graphed above in slope-intercept form.

$$y < \frac{2}{3}x + 2$$

2. Explain how to determine which inequality symbol to use when writing an inequality in slope-intercept form.

- The shading is below the line so when it is below the sign is $<$.
- You can also test out a point to check for accuracy.
- Since the boundary line is dotted, it must be $<$ or $>$.

3. Write the inequality of the solution set graphed above in point-slope form.

Answers may vary.

$$y - 4 < \frac{2}{3}(x - 3)$$

4. Explain how to determine which inequality symbol to use when writing an inequality in point-slope form.

Test a point or based on shading it is $<$ for below and $>$ for above.

5. Write the inequality of the solution set graphed above in standard form.

$$\begin{array}{lcl} y < \frac{2}{3}x + 2 & & -2 < \frac{2}{3}x - y \\ -2 & -2 & 3(-2 < \frac{2}{3}x - y) \\ y - 2 < \frac{2}{3}x & & -6 < 2x - 3y \\ -y & -y & = 2x - 3y > -6 \end{array}$$

Five inequalities are given. Complete the table below to match each inequality to the graph of its solution set by writing the inequality on the line provided under each graph.

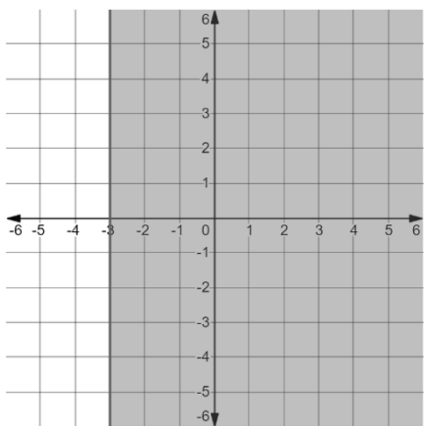
$$x \geq -3$$

$$y \leq -3x + 1$$

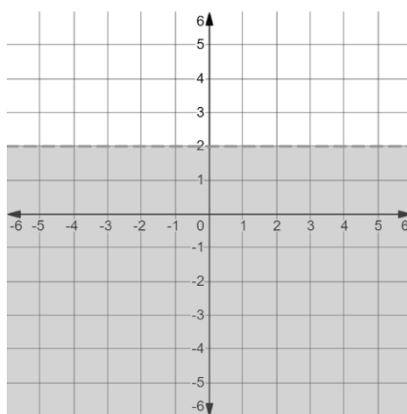
$$x - 2y > -6$$

$$y < 2$$

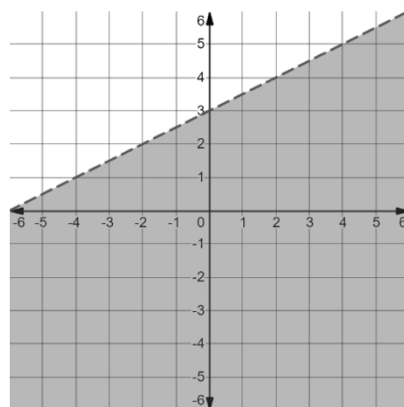
$$y + 1 > -\frac{3}{4}(x + 2)$$



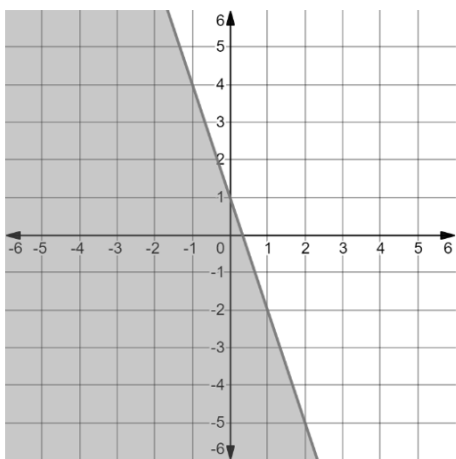
6. Inequality: $x \geq -3$



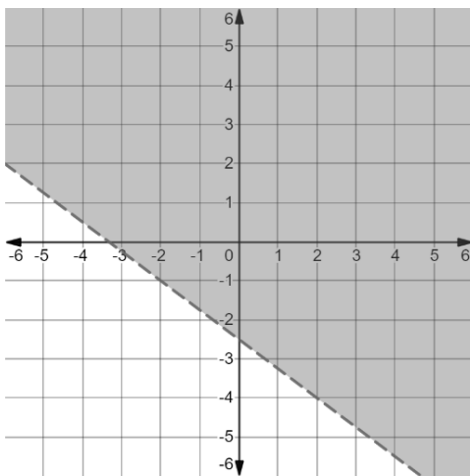
7. Inequality: $y < 2$



8. Inequality: $x - 2y > -6$



9. Inequality: $y \leq -3x + 1$



10. Inequality: $y + 1 > -\frac{3}{4}(x + 2)$